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SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

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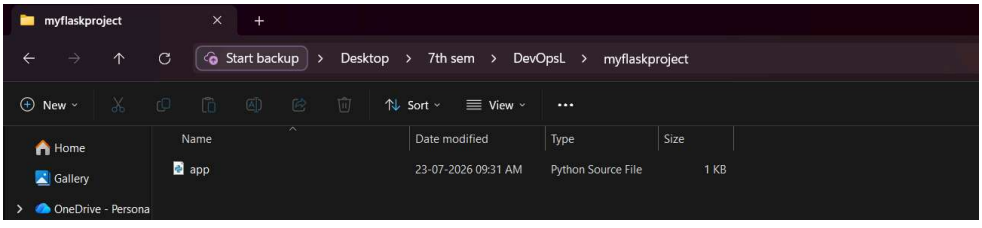
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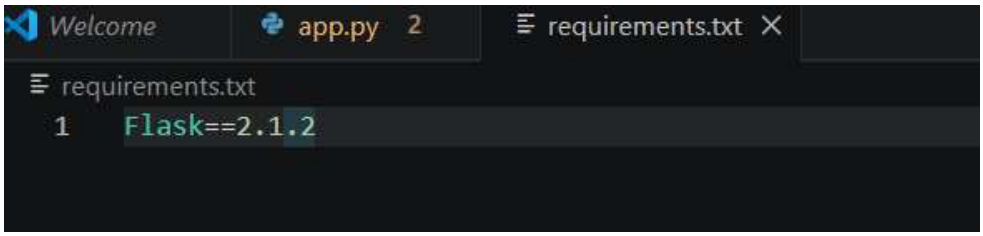
Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Assignment No. 08

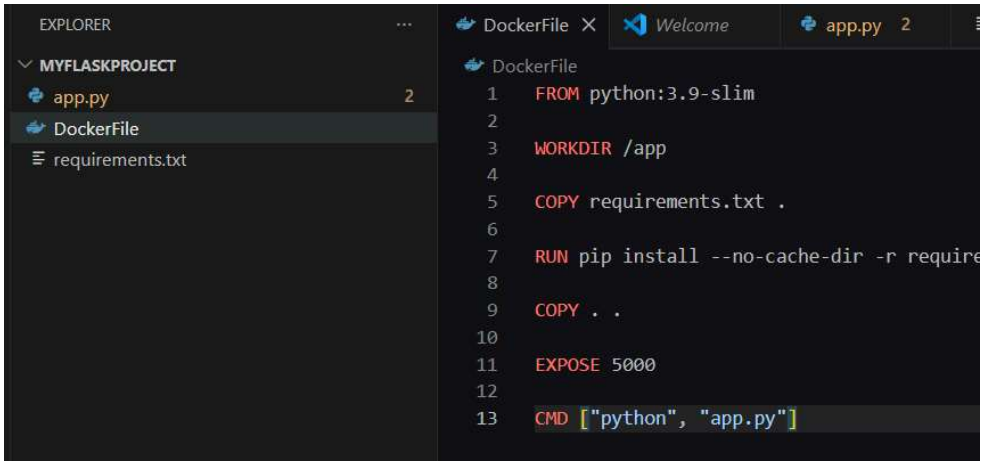
Subject:	DevOps Lab
TCODE:	TE7950
Name of Student	**Raghav Sharma**
PRN No.	**23070122171**
Branch	CSE, Batch (2023-27), L3
Academic Year & Semester	25 – 26, Sem VII
Date of Performance	23rd July, 2026
Title of Assignment:	Containerization with Docker Dockerfile &Image Build
Theory:	Docker is an open-source containerization platform that packages an application along with its dependencies into lightweight, portable containers. A **Dockerfile** defines the instructions to build a Docker image, ensuring consistent deployment across different environments. Containers created from Docker images provide isolated, scalable, and efficient execution of applications.

Implementation:	Step 1: Install Docker Desktop and verify the installation using `docker --version`.
	`C:\Users\sharm\Desktop\7th sem\DevOpsL\myflaskproject>docker --version`
	`Docker version 29.3.1, build c2be9cc`
	Step 2: Create a folder named `my-flask-app` and create `app.py` with the Flask Hello World application.
	 

Step 3: Create `requirements.txt` and add the Flask dependency.



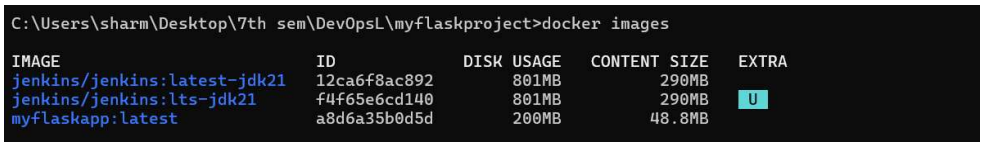
Step 4: Create a `Dockerfile` using the Python base image, copy application files, install dependencies, expose port 5000 and define the startup command.



Step 5: Open the terminal inside the project folder and build the Docker image using:
docker build -t my-flask-app .



Step 6: Verify the image using:
docker images



Step 7: Run the Docker container:
docker run -p 5000:5000 my-flask-app



Step 8: Open http://localhost:5000 in the browser and verify the "Hello, Docker!" output.

	 Hello, Docker!

Subject Faculty Name: Prof. Arijit Dutta